Core Diagnostic Engine

Core Diagnostic Engine User Guide

Diagnosing and Repairing Windows PCs

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Introduction

What Core Diagnostic Engine does

Core Diagnostic Engine (CDE) is a repair and diagnostics toolkit for Windows PCs. It helps you find problems (slow performance, malware, boot failures, update errors) and run trusted repair steps from one organized program.

CDE can run in two main environments:

Environment	When you use it	What it means
WinPE Mode	Boot from the CDE USB stick	A minimal Windows recovery environment. Best when Windows will not start normally or you need offline access to the customer's drive.
Windows Mode	Run CDE.exe on a working Windows install	Full access to live Windows services, Windows Update, and most repair tools while Windows is running.

What is on the USB toolkit

The CDE USB drive typically includes:

- **CDE WinPE** — boots into the graphical CDE program
- **Core Hardware Utility (CHU)** — pre-boot hardware tests before Windows loads
- **Windows Memory Diagnostic** — memory test tool (when staged on the USB)

How this guide is organized

Each chapter explains **what a tool does**, **when to use it**, and **step-by-step instructions**. Callout boxes highlight tips, warnings, and WinPE-only features.

Start with **Getting Started** if you are new to CDE, then follow the **Recommended Repair Workflow** for a practical order of operations on a typical service bench job.

Getting Started

Booting from the CDE USB

1. Insert the CDE USB drive into the computer.
2. Power on and open the **UEFI/BIOS boot menu** (often F12, F11, Esc, or F8 — varies by manufacturer).
3. Choose the USB drive as the boot device.
4. When the **rEFInd** graphical menu appears, select **CDE WinPE** to start the toolkit.

Tip: If Secure Boot blocks CHU or custom boot files, disable Secure Boot in firmware settings for diagnostic sessions.

WinPE boot menu (keys 1–7)

After WinPE loads, CDE shows a full-screen boot menu. Press a number key or click an option:

Key	Option	Use when
1	Launch CDE Toolkit GUI	You want CDE tools in WinPE without loading the customer's Windows registry
2	Launch CDE + Load Offline OS	You need to scan or repair the customer's installed Windows (malware scan, offline DISM, startup audit, account recovery)
3	WinPE Command Prompt	You need a raw command line
4	PowerShell (WinPE)	You need PowerShell scripts in WinPE
5	Network Setup	You need network in WinPE (wpeinit)
6	Reboot	Restart the computer
7	Shutdown	Power off the computer

Important: For most repair jobs on a customer's drive, choose option 2 so CDE can load their offline Windows installation. You will be asked to confirm before registry hives are loaded.

Tip: Options 1 and 2 continue in the same ToolkitGUI.exe process — the boot menu closes and the main CDE window opens. If you return to a text recovery menu instead, sync the latest GUI from Publish-ToolkitGUI.ps1 and check CDE\Runtime\winpe_startup.log on the USB.

Administrator access (UAC)

CDE runs with administrator rights. Windows will show a **User Account Control** prompt — choose **Yes**. Many repair tools need elevated access to system files, the registry, and disk volumes.

The CDE main window

After the boot menu (or when launching `CDE.exe` on a full Windows system):

- **Left sidebar** — tools grouped as Main, Repair, Maintenance, and Utilities
- **Top header** — computer name, environment badge (WinPE or Windows), CPU/RAM/SMART summary, uptime
- **Work order strip** — optional case tracking from the Dashboard

Work orders (Dashboard)

What this does: Keeps client name, phone, notes, and case ID with your session so diagnostics, repairs, and standalone scans are logged together.

1. Open **Dashboard** in the sidebar.
2. Enter client details and notes.
3. Click **Generate** or **Save** to start or update a work order.
4. Run diagnostics and repairs from other sidebar tools — results are tied to the active case.

Case timeline (active work order only) — newest-first list of workflow runs, CHU pre-boot result loads, and Process Analyzer CST scans. Timestamps show as local friendly times (for example `Jul 20, 2026 9:48 AM`). Use **Open Report** (PDF) or **Open artifact** on each row when paths are available.

Suggested path — under **What are you seeing?**, CDE highlights the best-matching symptom when you **Generate Work Order** (from notes keywords) or when you click a symptom chip. The hint text and accent highlight show the recommended path; the detail panel lists follow-up tools and health-check presets. Click a symptom to confirm or change the path.

Recent work orders — lists saved cases from `CDE\Runtime\WorkOrders\`. Click a case for run history in the detail window (Opened/Closed times use local friendly dates such as `Jul 20, 2026 9:48 AM`). Use **Clear Recent** to remove all case JSON files (closed-case PDFs in `WorkOrders\` are kept).

Symptom cards — five paths: Slow PC, Suspected Malware, Tech Support Scam, Network Issues, Won't Boot. Selecting one saves `GuidanceContext` on the work order and can jump you to the matching health preset or repair tool.

> Tip: On a USB toolkit, work orders are stored on the USB media so you can review past jobs later. Internal case JSON lives under CDE\Runtime\WorkOrders\; the printable PDF for a closed case is saved to WorkOrders\{CaseId}.pdf. Open a closed case from Recent work orders for run history, or use Open Final Report for the PDF.

Recommended Repair Workflow

Start here: a practical bench order

Use this order for a typical "computer won't behave" job. Adjust based on the customer's symptoms.

On the **Dashboard**, the **What are you seeing?** symptom cards mirror this workflow. Enter client notes and **Generate Work Order** — CDE infers the best path from keywords in your notes (or click a symptom chip to choose explicitly). The suggested card highlights immediately and shows follow-up tools.

```
Symptoms → Health Check → Targeted tests → Repair → Verify → Cleanup
```

Step 1 — Understand the problem

Ask what changed: updates, new software, malware pop-ups, no boot, slow performance, network loss. Note whether Windows starts normally.

Step 2 — Boot CDE correctly

Situation	Boot choice
Windows starts; run CDE on the live system	Launch CDE.exe as administrator
Windows won't start; repair from USB	USB boot → WinPE menu option 2 (Offline OS)
Hardware suspicion before OS loads	rEFInd → Core Hardware Utility (CHU) first

Step 3 — Run System Health Check

Open **System Health Check** → choose a preset (**Recommended**, **Quick Vitals**, **Deep System**, or **Full Audit**) or pick checks manually → **Run Health Check**.

Review the scored findings and the full technical detail beneath them. Yellow (Warn) and red (Fail) items tell you where to focus next; Severity, Confidence, Evidence, and Recommendations explain why. The selected card is marked **VIEWING** in the left list.

Step 4 — Deepen diagnostics as needed

If health check suggests…	Go to…
Disk problems	S.M.A.R.T. Analyzer and/or Diag & Repair → Storage
CPU/RAM stress	Diag & Repair → CPU / Memory
Malware concern	Malware Cleanup (Defender + Core Security Tool)
Browser/search hijacks, proxy/DNS tampering, security settings	Hijack & Scam Repair
Boot/startup issues	Startup Manager; security audits in Health Check
OS file corruption	OS Repair (SFC/DISM)

Step 5 — Apply repairs

Use the smallest fix that addresses the finding:

- **Fast Fixes** for quick network/cache/service issues on live Windows
- **OS Repair** for system file and component store problems
- **Disk & Boot Repair** (WinPE only) for boot manager and BCD issues
- **Post-Op Cleanup** after malware removal or major changes
- **Hijack & Scam Repair** when health checks or symptoms point to browser, network, or security-setting hijacks

Step 6 — Verify and document

Re-run critical checks. On the **Dashboard**, follow the highlighted **suggested path** under **What are you seeing?** — click the symptom card to open System Health with the matching preset (or use the follow-up buttons). The **Case timeline** card lists workflows, CHU pre-boot loads, and standalone CST scans for the active case.

Use **Open Report** from System Health Check, Hijack & Scam Repair, Process Analyzer, S.M.A.R.T. Analyzer, the workflow execution window (when shown), or the case timeline when you need a customer-facing PDF (`Logs\`). For **Diag & Repair** and **Disk & Boot Repair**, use the case timeline or browse `Logs\` directly. When starting a fresh case on a shared USB, clear scan artifacts from **Settings → Clear Test & Scan Logs**, then remove old case records from **Dashboard → Clear Recent** if the recent list is cluttered.

> WinPE only: Disk & Boot Repair, offline DISM/SFC, and offline Windows Update package removal require WinPE with the customer's OS loaded (menu option 2).

Diagnostics

This chapter covers the main **find problems** tools in the MAIN sidebar section.

System Health Check

What this does: Runs a catalog of automated checks (vitals, deep system info, audits, security reviews) and presents scored findings together with detailed technical inventories and a PDF report. On the results screen, you can drag-select and copy text from the REVIEW pane (evidence, impact, recommendations, and detailed output).

When to use it

- First pass on any machine (quick overview)
- Before and after major repairs (before/after comparison)
- When you need one report covering many areas

Steps

1. Sidebar → **System Health Check**
2. Use a preset (**Recommended**, **Quick Vitals**, **Deep System**, **Full Audit**) or tick individual checks in the category list (**Select All** / **Clear All** as needed)
3. Click **Run Health Check**
4. Review the left-side cards — green Pass, yellow Warn, red Fail. The currently open card has a blue highlight, left accent, and **VIEWING** badge
5. In the right panel, review:
   - **Actionable / verified findings** — Finding, Impact, Severity, deterministic Confidence (0–100 match score), Evidence, and Recommendations
   - **Detailed results** — the full adapter, device, service, event, or inventory information collected by the check
6. The **Findings** list follows the catalog order in `Config/HealthCheckCatalog.json` (`findingsDisplayOrder`: Activation through Remote Access)
7. Click **Open Report** for a printable PDF summary (`Logs\`)

Confidence: This is a deterministic rule/evidence match score from 0–100, not a probability that hardware or software is faulty.

What is included

Category	Examples
Quick Vitals	OS version, CPU load, RAM, disk space, activation, recent errors
Deep System	Driver signatures and filter inventory, Windows Update stack (idle Stopped + Manual wuauserv is normal on Win10/11), network adapters, GPU information, GPU functional tests, critical events
System Audit	Installed software, services, startup entries, event logs, partitions
Security Audits	Autoruns, hosts file, browser extensions, scheduled tasks, browser/network hijack, Defender exclusions and security tampering, ransomware recovery, miner behavior, remote access

Important: Heavy scans (SMART deep tests, Defender, line-item hardware tests) are not in Health Check — use S.M.A.R.T. Analyzer, Diag & Repair, and Malware Cleanup instead.

Event and GPU findings

- Event-log checks correlate related records into findings such as unexpected shutdown, BSOD, application crash, storage/NTFS, WHEA hardware, service, update, and display-driver failures. Raw/deduplicated event evidence remains available in the detailed output.
- **GPU Information** is inventory-only. **GPU Functional Tests** is a separate active check covering driver state, DirectX, OpenGL, Vulkan, a capped VRAM write/readback test, and best-effort temperature reporting.
- WinPE, Remote Desktop, Microsoft Basic Display, missing graphics runtimes, or unavailable vendor temperature tools can produce **N/A**. That means the test could not run in that environment; it is not automatically a failure.

S.M.A.R.T. Analyzer

What this does: Reads disk self-monitoring data (SMART) for fixed internal and external drives and shows health attributes.

Removable media: USB flash drives, memory cards, and other removable media are not listed or scanned (including the CDE boot USB in WinPE). USB external hard drives that report as fixed media may still appear.

Steps

1. Sidebar → **S.M.A.R.T. Analyzer**
2. Select a drive from the list

3. Click **Scan** (or refresh)
4. Review overall health badge and attribute table
5. When the scan finishes, **Open Report** enables and opens the PDF under `Logs\` (built from the scan results — status should not stay on "Saving…" forever)

Next steps

- Failing SMART → recommend drive replacement; run **Diag & Repair → Storage** for read tests
- Healthy SMART but slow disk → check free space (Health Check) and temp files (**Temp File Cleaner**)

Diag & Repair

What this does: Runs detailed line-by-line hardware tests (CPU, memory, storage) plus an optional **Core Security Tool** scan-only pass — similar to commercial PC diagnostic suites.

Profiles

Profile	Use when
Quick	Bench triage; shorter run times
Extended	Deeper stress; more thorough disk and CPU work

Steps

1. Sidebar → **Diag & Repair**
2. Choose **Quick** or **Extended**
3. Click **Run All Tests** or run **CPU Tests**, **Memory Tests**, **Storage Tests**, or **CST Scan** individually
4. Watch the live results table as each sub-test completes
5. After CST scan, review the post-scan dialog for detections (remediation optional)
6. A PDF report is saved under `Logs\` when the run finishes. With an active work order, use **Dashboard → Case timeline → Open Report**; there is no **Open Report** button on the Diag & Repair screen itself

> Tip: Run CDE as Administrator for full storage physical-read tests. In WinPE with offline Windows loaded, CST can scan the customer's volume.

> Removable media: Storage tests run only on fixed testable drives. USB flash sticks, SD/microSD cards, and the toolkit boot USB are excluded automatically. If no fixed drives are present, storage discovery fails with no physical drives detected.

CPU tests (PowerShell 5.1)

On Windows 10/11, CDE runs under **PowerShell 5.1**. CPU SIMD line tests (MMX, SSE, SSE2, …) use **CdeCpuSimdCli.exe**, published self-contained next to `ToolkitGUI.exe`. You should see **Pass** or **Fail** per row — not a permanent **Warn** with "stub engine." If every SIMD row warns about a stub, republish with `Publish-ToolkitGUI.ps1` and run `SyncUSB.bat`.

Testable drive policy (Diag & Repair, S.M.A.R.T., CHU)

CDE and CHU use the same intent: **do not hardware-test removable media.**

Surface	What is excluded	What may still be tested
Diag & Repair → Storage	Removable media (Win32_DiskDrive.MediaType / Get-Disk.IsRemovable)	Internal SATA/NVMe; many USB external HDDs (fixed media)
S.M.A.R.T. Analyzer	Same filter via PhysicalDiskCatalog.psm1	Same
CHU (UEFI)	EFI_BLOCK_IO_MEDIA.RemovableMedia at drive enumeration	Fixed disks only in the storage test tree

Not affected: Disk & Boot Repair still lists all disks (including removable) for layout and repair work. System Health disk inventory is unchanged.

CHU Pre-Boot Results

What this does: Shows JSON results from **Core Hardware Utility** (`CoreDiag.efi`) after you run pre-boot hardware tests from the rEFInd menu.

Steps

1. Boot USB → rEFInd → **Core Hardware Utility** (loads `CHUPrep.efi`, then `CoreDiag.efi`)
2. Run memory, storage, or CPU tests (or Run All) — storage tests target **fixed** drives only (boot USB and other removable media are not listed)
3. After tests, use **UP/DOWN** in the results review screen to scroll the full list; press **ENTER** to return to the CHU menu
4. Boot back into CDE WinPE or Windows
5. Sidebar → **CHU Pre-Boot Results** → **Refresh**

Results are stored on the toolkit volume under `EFI\CoreDiag\Results`. Session logs are under `EFI\CoreDiag\Logs` (written from early boot when the `CDEDATA` volume is available).

> Display modes: On many desktops CHU uses a PC-Check-style text interface. On some laptops the firmware shows only a GOP framebuffer — CHU switches to a graphics menu automatically. Both paths support the same tests and review scroll.

The status bar shows a **rollup** (pass / warn / fail counts). When a test maps to the hardware catalog, the row shows the catalog display name. For warned or failed tests, use **Open matching Diag & Repair test** to jump to the same test in Diag & Repair (highlighted, no auto-run). With an active work order, loading results also adds a **CHU pre-boot** entry to the Dashboard case timeline.

CHU vs Diag & Repair

Tool	When it runs	Best for
CHU (UEFI)	Before Windows starts	Raw hardware validation, no OS interference
Diag & Repair (WinPE/Windows)	Inside CDE GUI	Technician workflow, combined with CST, live progress in CDE

Use both when a machine has intermittent hardware faults — CHU for pre-boot proof, Diag & Repair for documented in-session runs with PDF workflow output.

When Diag & Repair reports **partial memory coverage**, **CPU stress warnings**, or **storage elevation warnings**, the view shows a CHU guidance banner with a link to **CHU Pre-Boot Results**. That means Windows-side coverage is incomplete for clearing intermittent hardware risk — run CHU from the USB pre-boot menu for physical validation.

Security

Malware Cleanup

What this does: Runs antivirus and custom malware scanning tools from one screen.

Tools in this view

Tool	Purpose
Windows Defender Scan	Microsoft's built-in quick scan (live Windows)
Core Security Tool (CST)	CDE's advanced scan engine with configurable scope and remediation

Steps — Core Security Tool

1. Sidebar → **Malware Cleanup**
2. Click **Open** on **Core Security Tool**
3. In the config dialog, set scan type (**ThreatScan** / **FullScan** / **CustomScan**), detection scope (**All** recommended), and optional custom paths
4. The **CST Scan Results** window opens immediately with live progress (same layout as File Manager / Startup Manager / Process Analyzer). Pre-scan steps (targets, known-good baselines, path discovery) and file analysis both update the progress panel — the window should not sit idle with no status
5. When finished, the same window shows the verdict, metrics, and primary findings. If review-only rows exist, a single collapsed **Suppressed audit Findings** line appears at the bottom of the Findings card; expand it to inspect those rows
6. Use **Copy summary** or **Open Report** for coverage notes and full scan narrative (coverage/action summary is not shown in the dialog footer)
7. Use **Open Report** to open the operator PDF (`Logs\CST_Scan_*.pdf`) generated when results appear
8. Select **actionable** findings and use **Quarantine Selected**, **Remove Selected**, or **Restore from Manifest...** — remediation runs with a **per-item progress** window, then a summary. No separate post-scan operator dialog is required after Malware Cleanup CST

9. **Archive hits:** inner `zip::entry` rows are review-only; select the **parent .zip/.jar** row to quarantine or remove the container

ThreatScan is a high-risk-path triage (not full-disk AV). It walks ProgramData, Temp, Tasks, Startup, Downloads, Users\Public, per-user Desktop / Documents / Roaming / INetCache (and more). Prefer FullScan or a folder CustomScan when hunting outside those roots. Authenticode timeout alone is a coverage event and does not create a security finding. An unsigned DLL in a risky path is only a weak signal unless another detector corroborates it; unsigned executables retain stronger handling. Empty EFI hash baselines are noted in Copy summary / PDF Coverage, not as a findings row.

WinPE + offline OS: Boot menu option 2 so CST can scan the customer's Windows folder. Without offline load, CST may show "no Windows volume found."

Update Hash IOCs

Threat signature updates are in the CST config dialog under **Threat Signature Database** (not a separate Malware Cleanup button).

Security audits vs malware scans

CDE separates **active scanning** from **configuration audits**:

Activity	Where to run it
Defender scan, CST malware scan	Malware Cleanup
Autoruns, hosts file, browser extensions, scheduled tasks, browser/network hijack, Defender exclusions/security tampering, ransomware recovery, driver/filter inventory, remote access	System Health Check → Security Audits or Deep System category

Scheduled Tasks Audit flags non-Microsoft tasks with suspicious execute paths or unsigned COM handlers. Microsoft-signed software under AppData (e.g. OneDrive) is not flagged for path alone. COM-only actions show the resolved handler (`COM {guid} -> path`) and a **Review Reason** when flagged.

Security Tampering Audit inventories risky Microsoft Defender exclusions when Defender is available. If an **active third-party antivirus** is registered in Windows Security Center (for example Malwarebytes), Defender disabled/passive state and Defender API errors such as `0x800106BA` are treated as expected coexistence — the Pass result names that product. Firewall, UAC, and SmartScreen weakenings still Warn. If Defender is disabled with **no** active third-party AV, those Defender findings remain Warns.

Ransomware Recovery Audit reviews VSS/backup-service and Windows Recovery Environment state for recovery impairment. **Driver Signature Audit** also inventories file-system minifilters and disk/volume UpperFilters/LowerFilters; names on the shared CST filter allowlist (stock Windows and known security-product filters) are not suspicious merely because they are present.

Important: Do not look for Autoruns or Hosts File checks under Malware Cleanup — they were moved to System Health Check only.

Hijack & Scam Repair

What this does: Scans for hijacked browser settings, network redirects, suspicious startup entries, tampered security settings, and remote-access tooling — then lets you repair selected items or review startup rows in Startup Manager.

Scan workflow

1. Sidebar → **Hijack & Scam Repair**
2. Click **Scan** (becomes **Re-Scan** after the first completed scan)
3. While scanning, the findings grid shows a centered **Scanning...** overlay (same pattern as Startup Manager loading)
4. Review flagged rows in the grid; select a row for **Finding Details** below the grid (location, current value, repair action, guidance). Drag the splitter between Findings and Finding Details to resize
5. Check **FIX** on repairable rows, or click **Select repairable** to check all flagged items that can be fixed here
6. Click **Repair Selected** for bulk fixes, or **Repair this item** in Finding Details for one row

Grid columns

Column	Meaning
FIX	Include in Repair Selected (enabled only when ACTION = Fix here)
ACTION	Fix here (auto-repair), Startup Manager (review-only startup row), or Info only
CATEGORY	Browser, Network, Security, Startup, Remote, etc.
RISK	Severity (e.g. Hijacked, Suspicious)
NAME	Short description of the finding

Reports and export

After a scan completes:

Button	Use for
Open Report	Open the operator PDF from Logs\ (generated automatically after the scan)

Use **Show scan log** in the detail toolbar if you need the raw workflow output.

Startup Manager integration

- **Startup Manager** toolbar button opens Startup Manager with **Suspicious only** enabled
- **Manage** on a startup finding jumps to Startup Manager filtered to that entry
- Startup Manager **Suspicious only** uses hijack-scanner heuristics for startup entries and service-specific risk heuristics on the Services tab

Quick Repairs drawer

Expand **Quick Repairs** at the bottom of the view for bulk scripts (browser, network, security settings, repair-all). These run full repair workflows and may require elevation. Quick Repairs sits in its own bottom row so it does **not** cover **Finding Details**.

> Tip: Scan results stay in memory when you switch to another sidebar tool and return — you do not need to re-scan to review findings. Run Re-Scan after making system changes.

> Note: Some startup and extension findings are review-only in this scanner; disable or delete them in Startup Manager rather than Repair Selected.

After malware removal

1. Re-run **System Health Check** security audits
2. Run **Post-Op Cleanup** (browser cache, services, file associations, DNS/proxy reset, icon cache, optional hosts restore)
3. Check **Startup Manager** for suspicious startup entries
4. Deliver a **PDF report** from Malware Cleanup or Health Check to document what was found

Repair

OS Repair

What this does: Repairs Windows system files, the component store, network stack, and Windows Update when the OS is damaged or misconfigured.

Tools

Tool	Full Windows	WinPE (offline OS loaded)
System File Checker (SFC)	Online scan	Runs DISM instead (SFC needs a running component store)
DISM	Online repair (optional install.wim source)	Offline repair against selected Windows folder
Network Stack Reset	Yes	Not available in WinPE
Windows Update Repair	Yes	Not available in WinPE
Offline Update Uninstall	No	WinPE only — remove bad updates from offline image

Steps — SFC / DISM (typical)

1. Sidebar → **OS Repair**
2. In WinPE, confirm the correct **offline Windows installation** is selected

3. Click **SFC** or **DISM** and wait for completion (may take 30+ minutes)
4. For **DISM** on full Windows, choose whether to browse for a matching **install.wim** / **install.esd** (recommended when Windows Update cannot supply repair files)
5. Review Pass/Fail/Warn in the workflow window; open PDF if needed
6. Reboot and test Windows normally

DISM error 0x800f081f (source files not found)

If DISM completes with **Warn** and reports **0x800f081f**, Windows Update could not download the files needed to repair the component store.

1. Mount a **Windows ISO** that matches the PC's edition and build (e.g. Windows 10 22H2 for build 19045.x)
2. Run **DISM Image Repair** again → **Yes** → select `sources\install.wim` from the mounted ISO
3. After DISM reports **Pass**, run **SFC** on full Windows
4. If online repair still fails, try **Windows Update Repair** in OS Repair, then rerun DISM

Manual elevated command (replace drive letter and index):

```
DISM /Online /Cleanup-Image /RestoreHealth /Source:wim:D:\sources\install.wim:6 /LimitAccess
```

Use `DISM /Get-WimInfo /WimFile:D:\sources\install.wim` to find the correct index.

> Important: Never assume C:\ is the customer drive in WinPE — always verify the offline Windows path.

Disk & Boot Repair

What this does: Fixes boot configuration (BCD), EFI boot files, and master boot record issues when Windows will not start.

> WinPE only: This entire section is disabled when running CDE on a normal Windows desktop. Boot from USB to use it.

Steps

1. Boot USB → WinPE menu option **1** or **2**
2. Sidebar → **Disk & Boot Repair**
3. Review the **read-only disk map**
4. Common actions:
   - **Backup BCD** before changes
   - **Scan OS** to find Windows installations
   - **BCD Repair** / **Rebuild BCD** / **Rebuild EFI** as appropriate
   - **Fix MBR** / **Fix Boot** on legacy BIOS systems if applicable

Always back up BCD stores before destructive boot repairs.

Fast Fixes

What this does: Fourteen one-click repairs with automatic Pass/Fail verification — ideal for quick bench fixes on a working Windows session.

Categories

Category	Examples
Networking	Reset Winsock, renew IP, disable IPv6
OS	Clear event logs
Services	Enable Windows Update, BITS, Defender
Browsers	Reset Edge/Chrome, fix browser file associations
Performance	Disable UI animations, clean Start Menu cache, high performance power plan

Note: DNS flush and icon/thumbnail cache clear live in Post-Op Cleanup only (removed from Fast Fixes 2026-06-26).

Steps

1. Sidebar → **Fast Fixes**
2. Filter by category if needed
3. Click a fix → confirm → wait for **Pass** or **Fail**
4. Read the log line if verification fails

Tip: Fast Fixes are for live Windows. Most show N/A in WinPE.

Maintenance and Utilities

Post-Op Cleanup

What this does: Post-malware and post-repair cleanup — browser caches, service defaults, file associations, DNS/proxy reset, shell caches, and optional hosts restore.

Tools

Tool	Use for
Browser Cache Clear	Edge/Chrome/Firefox disk cache (not cookies/passwords)
Restore Default Services	Reset critical Windows services to default start types
File Association Fix	Repair hijacked .exe, .lnk, .bat, .cmd, .com handlers

Tool	Use for
Icon + Thumbnail Cache Clear	Fix broken or stale desktop icons and thumbnails
DNS Cache Flush	Clear resolver cache after DNS hijack or VPN issues
Reset Proxy Settings	Disable user Internet proxy and reset WinHTTP proxy
Restore Hosts File	Replace the live Windows hosts file with defaults (backup first; confirmation required)
Restart Explorer	Apply shell/cache fixes without a full reboot

> Note: The Hosts File Check in System Health Check still audits the hosts file; remediation lives here only.

Restore Hosts File targets the active `%SystemRoot%`, clears a read-only attribute when present, and requires elevation to write the protected file. If the operation fails, the result includes the underlying path or permission error instead of only a generic failed-step status.

Steps

1. Sidebar → **Post-Op Cleanup**
2. Click **Run** on each tool you need (eight tools; no single **Run All** button)
3. Approve UAC if prompted (services, associations, and hosts restore need elevation on live Windows)

Backup & Imaging

What this does: Copies files and folders with Robocopy — mirror or backup mode with optional dry run.

Steps

1. Sidebar → **Backup & Imaging**
2. Click **Robocopy Backup**
3. Set source, destination, mirror mode, and dry-run if testing
4. Run and review the log summary

> Tip: Use File Manager → Shadow Copies (full Windows) to browse previous versions read-only before backup decisions.

Startup Manager

What this does: Lists and controls programs that start with Windows — Run keys (including **GPO** policies), startup folders, **MSIX startup tasks** (Store apps such as Teams), **scheduled logon/boot tasks**, services, drivers, browser extensions, and Winlogon entries. CDE aligns enable/disable behavior with **Task Manager** via the `StartupApproved` registry hive where Windows uses it.

Tabs

Tab	Use for
User logon	Run keys, GPO Run keys, startup folders, MSIX tasks, scheduled tasks, Userinit/Shell reset when hijacked
Drivers	Boot-start drivers
Extensions	Browser extensions and legacy BHO entries
Services	Review service risk, filter suspicious services, change startup type, start/stop/restart, and delete eligible services

Persistence types (User logon)

Source	Notes
Run / RunOnce	HKLM and HKCU; CDE moves disabled values to AutorunsDisabled and syncs StartupApproved
GPO Run	Policies\Explorer\Run — listed and managed like standard Run keys
Startup folder	Active folder vs CDE\DisabledStartup; StartupApproved synced for Task Manager parity
MSIX startup task	Per-package tasks under AppModel (e.g. Teams, Copilot); disable via package State DWORD. CDE resolves the startup executable from the Appx manifest (windows.startupTask TaskId) and checks Authenticode when the file is accessible. If WindowsApps blocks direct file access, Signed shows Package (Publisher) instead of Not found.
Scheduled task	Tasks with logon, boot, registration, or session triggers; enable/disable via Task Scheduler. Non-Microsoft tasks can be Delete'd (unregister). Windows \Microsoft\ tasks are hidden by default (checkbox: Include Windows scheduled tasks); when shown, they are read-only.

When the same app appears through more than one mechanism, the **RELATED** column and analysis tooltip warn you (e.g. Teams via MSIX task and a Run key).

Right-click actions

Right-click a row for technician jump tools (Enable / Disable / Delete stay on the row buttons only):

Action	Use for
Open target folder	Explorer select on the resolved target (prefers -File scripts over powershell.exe)
Copy target file path / command line	Paste into notes or another tool
Delete target file...	Remove a leftover dropper under ProgramData / AppData / Temp / Public (not OS install paths). Does not remove the autorun itself

Action	Use for
Scan with CST	Startup-entry scan: multi-target file CustomScan + command-line LOLBin analysis + persistence correlation; result dialog shows File / Command line / Persistence verdicts
Copy registry path / Open in Registry Editor	When the entry has a registry location
Show entry details	Full name, source, risk, analysis, command line, resolved target, related
Open Malware Cleanup / Process Analyzer / File Manager	Jump to related workflows

When you **Delete** a Run/Startup/**non-Microsoft scheduled task** entry whose target still exists in a remediable path (e.g. `ProgramData\VendorCache\...`), CDE asks whether to **also delete that target file** (and empty parent folders). Prefer Yes for leftover malware droppers after clearing persistence.

Services risk and delete

The Services tab assigns a **RISK** level and analysis from the service image path, signature/publisher, launcher usage, missing or risky locations, remote-access naming, and other persistence indicators. **Suspicious only** applies to this tab and shows only services that match those service-specific heuristics.

Eligible suspicious or unprotected third-party services show **Delete** on the row and right-click menu. Delete asks for confirmation, stops the service when necessary, and permanently removes its Windows service registration. It **does not delete the service executable**.

CDE does not offer service Delete for Windows service hosts (`svchost.exe` / `services.exe`), protected Windows services, kernel or driver services, ordinary services whose binary is under the Windows folder, or offline Windows sessions. Manage drivers from the **Drivers** tab.

Services also offer the same file/CST/copy/details jumps as other rows, plus **Copy service name**.

> Tip: Use System Health Check → Scheduled Tasks (security audit) to review non-Microsoft scheduled tasks for persistence risk. Flagged rows include a Review Reason; COM-only actions show the resolved handler path (not "unknown action"). Microsoft-signed tasks under AppData (e.g. OneDrive) are not flagged solely for path location. The default Startup Manager view focuses on user-facing startup mechanisms like Task Manager.

Winlogon reset

If `Userinit` or `Shell` values were changed by malware, they appear highlighted. Use **Reset** to restore Windows defaults (backup created automatically). **Undo** restores the previous value if reset was wrong.

Hijack & Scam Repair integration

- **Suspicious only** filter — shows startup entries that match hijack-scanner heuristics and Services that match service-specific risk heuristics
- **Manage** from a hijack finding opens Startup Manager with this filter enabled
- Review-only startup findings in Hijack & Scam Repair should be disabled or deleted here rather than via **Repair Selected**

File Manager

What this does: Browse drives and folders in a split-pane layout — useful for copying logs, pulling user data, or inspecting offline OS files.

Steps

1. Sidebar → **File Manager**
2. Expand **Computer** → select a **drive** or **shadow copy**
3. The right pane shows folder contents
4. Use checkboxes to select multiple files or folders for batch copy, move, or delete
5. Right-click for Open, Cut/Copy/Paste, Rename, Delete, Properties, Show in Explorer, **Scan with CST**, or New folder
6. **Scan with CST** — right-click a file or folder (or multi-select) to run a Core Security Tool CustomScan on those paths. Empty area of the pane: **Scan this folder with CST** scans the current folder. Results open in **CST Scan Results** (live progress → findings). Archive threats: select the parent `.zip/.jar` row to quarantine/remove; inner entry rows are review-only.

The directories pane starts empty until you select a location.

> Note: Junction/reparse-point folders (e.g. C:\ProgramData\Application Data) are hidden to prevent infinite navigation loops. Folder size calculations skip the same paths.

Process Analyzer

What this does: Live process triage on the running system — CPU and RAM usage, image path, risk heuristics, network activity, and a detail pane for command line, session, parent chain, loaded modules, module anomalies, connections, and file hash. Use during malware investigations or when System Health Check flags high CPU or memory.

Main controls

Control	Use for
Refresh	Reload the process list
Pause / Resume refresh	Stop or resume auto-refresh (default ~10s)
Freeze	Hold the current snapshot while you inspect rows
Investigation ON	3-second refresh; highlights processes that started after you enabled the mode. The button label shows when investigation is active (no separate status banner in the toolbar).
Open Report	Open the operator PDF for the current filtered process snapshot (Logs\)
Terminate	End the selected process (use with care)

Filters and views

- Search box — filter by name, path, company, PID, or risk flags
- Preset dropdown — e.g. high CPU (≥10%), high RAM (≥500 MB), unsigned (with **Deep scan**), suspicious path, in

startup, **New since investigation**

- **Tree view** — parent/child process tree
- **Deep scan (signatures)** — Authenticode check on image paths (slower refresh)

Detail pane (select a row)

Shows command line, parent PID, **session** (0 = services/system; 1+ = user desktop), start time, signature, RAM, parent chain, top loaded modules, SHA256, and:

- **Network connections** — protocol, local/remote endpoints, state, and owning PID when available
- **Module anomalies** — loaded-module conditions that merit review
- **Risk flags** — credential-access, injection, miner, suspicious command-line/path, and related indicators; these are triage signals, not automatic proof of malware

Action buttons:

Button	Action
Open location	Open the process image folder in Explorer
Copy path	Copy image path to clipboard
CST scan file	Run Core Security Tool against the selected executable only; results open in a structured dialog
Startup Manager	Jump to Startup Manager
Malware Cleanup	Jump to Malware Cleanup

Note: Many Windows services legitimately show Session: 0 and an empty command line. Interactive apps such as explorer or your browser should show Session: 1 (or higher on RDP) and a populated command line.

Tip — svchost rows: Use the detail pane Command line to see the -k service group and -s service name (e.g. -k netsvcs -s wuauserv). The Services (svchost) list names hosted services when available. Session 0 and C:\Windows\System32\svchost.exe without -s usually indicate a shared service host, not a standalone app.

From System Health Check

When CPU or RAM checks warn or fail, use the recommendation link **Process Analyzer (high CPU)** or **(high RAM)** to open this view with the matching filter already applied.

Temp File Cleaner

What this does: Finds temporary files across user profiles and system temp locations, grouped by folder, for safe cleanup.

1. Choose drive scope
2. **Scan**
3. Review sizes and select folders

4. **Clean** selected items

Completion status

A successful clean may still leave some files behind when another process has them open — this is normal on a running system. CDE treats these as **skipped (in use)**, not failures:

- Files reported as *being used by another process*
- Files where delete returns *access denied* (common for `VBCSCompiler` / Roslyn analyzer DLLs while an IDE or `dotnet build` is active)

When only these expected skips occur, the workflow reports **Completed** / **Pass** — not a warning. The summary may note how many items were skipped (for example, `In-use: 5`). Genuine delete errors (unexpected I/O problems) still show as warnings.

> Tip: Close heavy build tools or reboot before cleaning if you want the temp folder as empty as possible.

Uninstall Manager

What this does: Lists installed programs from registry uninstall keys. Run a preferred uninstaller or force-remove broken ARP entries.

Uninstall: Prefers `QuietUninstallString` when present; otherwise normalizes MSI product codes to `msiexec /x {GUID} /qn`; otherwise runs `UninstallString`. After the uninstaller exits, CDE checks whether the ARP key and install folder remain and can offer safe leftover-folder cleanup (Program Files / ProgramData only).

Force Remove: Deletes the uninstall registry (ARP) key so the entry disappears from Programs and Features. Optionally deletes the recorded install folder, then offers a confirmed leftover-folder scan. This is not a full product wipe by itself — use it for broken or spoofed entries when no real uninstaller exists.

Tools Library and Settings

Tools Library

What this does: Launches onboard boot tools, portable apps, and special CDE workflows from a catalog.

Common entries

Category	Tool	Notes
Boot Environment	Core Hardware Utility	rEFInd loads CHUPrep.efi, which chainloads CoreDiag.efi
Boot Environment	Windows Memory Diagnostic	Memory test EFI payload
Recovery & Network	Firefox ESR	Portable browser (Ethernet in WinPE)
Account Recovery	Enable Offline Administrator	WinPE + offline OS only
Account Recovery	Disable Offline Administrator	Reverses offline admin enable

Steps — boot tool

1. Sidebar → **Tools Library**
2. Find the tool card
3. Click **Launch** or **Run**

> WinPE only: Account recovery tools require boot menu option 2 and a detected Windows installation.

Settings

What this does: Toolkit preferences, power actions, log cleanup, and this user guide.

About

Shows CDE version information.

Power (WinPE)

Button	Action
Unmount OS	Unloads offline registry hives without rebooting
Reboot Windows	Unloads hives then reboots the PC

Use **Unmount OS** before removing the USB if you loaded an offline OS and want to return to the boot menu without restarting.

Diagnostic logs

Clear Test & Scan Logs removes:

- `EFI\CoreDiag\Logs` — CHU UEFI logs
- `EFI\CoreDiag\Results` — CHU JSON results
- `Logs\` — operator PDF deliverables
- `CDE\Runtime\Data\` — internal workflow/scan/CST JSON and PowerShell session logs

Work order case JSON under `CDE\Runtime\WorkOrders\` is **not** removed here. Use **Dashboard → Recent work orders → Clear Recent** to delete saved case records (closed-case PDFs in `WorkOrders\` are kept).

Confirm before deleting — this cannot be undone.

Help

Open User Guide opens this PDF from `HALP\CDE_User_Guide.pdf` on the toolkit media.

Launching CDE on full Windows

Outside WinPE, run the canonical launcher:

```
Toolkit\CDE.exe
```

`CDE.exe` sets `TOOLKIT_ROOT` and starts the graphical toolkit. Always approve the administrator prompt.

Understanding Reports

Where reports are saved

CDE separates **operator deliverables** from **internal app data**:

Location	Contents
Logs\	PDF only — printable reports for customers and case files
CDE\Runtime\Data\workflows\	Internal workflow JSON (step details, health summaries, structured findings and actions)
CDE\Runtime\Data\scans\	Internal JSON for Hijack & Scam Repair and Process Analyzer snapshots
CDE\Runtime\Data\cst\	Internal CST scan JSON, plus correlation artifacts such as CST_BehaviorTelemetry_* and CST_NetworkSnapshot_*
CDE\Runtime\Data\sessions\	PowerShell session .log files (not written to Logs\)

CHU pre-boot tests write separately to **EFI\CoreDiag\Logs** and **EFI\CoreDiag\Results**. Logs are opened early in boot (before the CHU UI initializes) when the `CDEDATA` volume is writable.

When a workflow or scan finishes, CDE generates the internal JSON first, then builds the matching PDF under `Logs\` with the same base filename.

Opening reports

After a workflow or scan finishes, use **Open Report** in the tool, workflow window, or case timeline. CDE opens the PDF from `Logs\` through Windows Explorer (works when CDE is elevated).

Tool	Where to open
Workflow execution window	Open Report (Artifacts panel) — shown for OS Repair, Malware Cleanup, Post-Op Cleanup, Fast Fixes, Temp File Cleaner, Backup & Imaging, and similar workflow tools
System Health Check	Open Report on the results screen
Hijack & Scam Repair	Open Report after a scan
Process Analyzer	Open Report after processes are loaded
S.M.A.R.T. Analyzer	Open Report after a scan
CST Scan Results	Open Report when live CST results finish (Malware Cleanup / File Manager / Startup Manager / Process Analyzer / Core Security Tool)

Tool	Where to open
Diag & Repair	Dashboard → Case timeline → Open Report when a work order is active, or open the matching .pdf in Logs\ (no in-view button; live progress stays on the Diag screen)
Disk & Boot Repair	Dashboard → Case timeline → Open Report, or browse Logs\

There is no **Save report** or **Open HTML** button in the GUI — operators use the PDF in `Logs\`. Internal JSON under `CDE\Runtime\Data\` is cleared with **Settings → Clear Test & Scan Logs**; work order case JSON is cleared separately with **Dashboard → Clear Recent**.

What PDF reports contain

- Workflow name and timestamp
- Overall health summary (Pass / Warn / Fail counts)
- Step-by-step results in plain language
- System Health PDFs label each check using friendly names from the health catalog
- System Health findings include Finding, Impact, Severity, deterministic Confidence (0–100 match score), Evidence, and Recommendations when available
- System Health keeps detailed legacy output (adapter/device/service/event inventories) alongside the scored findings; older text-only reports still use the legacy formatter
- Hijack & Scam Repair and Process Analyzer PDFs use a findings/process table layout (landscape) suitable for bench notes
- CST PDFs retain primary findings and review-only suppressed audit rows. In the live CST window, suppressed rows are collapsed under **Suppressed audit Findings** until expanded

Work orders and logging

When a work order is active on the Dashboard, runs are appended to that case:

Event kind	Typical source
Workflow	Malware Cleanup, OS Repair, Diag & Repair, Post-Op Cleanup, Temp File Cleaner, Fast Fixes, etc.
ChuPreBoot	Refresh on CHU Pre-Boot Results (once per JSON file)
CstScan	Process Analyzer CST scan file

Each run may store paths to internal JSON and the operator PDF. The Dashboard **Case timeline** and work order detail window provide **Open Report** when the PDF still exists on disk.

Use consistent case IDs for shop billing and revisit history on the USB.

Clearing old reports

Before a new customer job on a shared USB:

1. Sidebar → **Settings**
2. **Clear Test & Scan Logs**

3. Confirm deletion

This removes CHU `EFI\CoreDiag\Logs` and `Results`, operator PDFs in `Logs\`, and internal data under `CDE\Runtime\Data\` (workflows, scans, CST, session logs).

To clear saved work order case records (not the closed-case PDFs), use **Dashboard → Recent work orders → Clear Recent**.

Locked files (active scan still running) may fail to delete — wait for the workflow to finish and try again.

Reference

Sidebar map

Section	Items
MAIN	Dashboard, System Health Check, S.M.A.R.T. Analyzer, Diag & Repair, CHU Pre-Boot Results
REPAIR	OS Repair, Malware Cleanup, Hijack & Scam Repair, Disk & Boot Repair
MAINTENANCE	Backup & Imaging, Post-Op Cleanup
UTILITIES	Tools Library, File Manager, Startup Manager, Process Analyzer, Temp File Cleaner, Uninstall Manager, Fast Fixes, Settings

WinPE capability matrix

Feature	WinPE (no offline)	WinPE + offline OS	Full Windows
System Health Check	Partial	Most audits	Full
Malware Cleanup (CST)	Limited	Full offline scan	Full live scan
Defender scan	N/A	N/A	Yes
OS Repair SFC	DISM path	Offline DISM	Online SFC
Disk & Boot Repair	Yes	Yes	Gated off
Fast Fixes	Mostly N/A	Mostly N/A	Full
File Manager shadow copies	No	Drive browse	Yes
Account recovery tools	No	Yes	N/A

Glossary

Term	Meaning
BCD	Boot Configuration Data — Windows boot settings database
CHU	Core Hardware Utility — UEFI pre-boot hardware diagnostics (CHUPrep.efi → CoreDiag.efi from rEFInd)
CdeCpuSimdCli	Small .NET helper beside published ToolkitGUI.exe; runs real CPU SIMD tests when CDE uses PowerShell 5.1
CdeGpuDiagCli	Self-contained .NET helper beside published ToolkitGUI.exe; runs System Health GPU probe/functional tests with environment-aware N/A results
CST	Core Security Tool — CDE malware scan engine
DISM	Deployment Image Servicing and Management — repairs Windows component store
Hive	A registry database file loaded offline in WinPE (e.g. customer's SOFTWARE)
Offline OS	Customer's Windows installation accessed from WinPE without booting it normally
rEFInd	UEFI boot menu on the CDE USB
SFC	System File Checker — verifies and repairs protected Windows files
SMART	Self-Monitoring Analysis and Reporting Technology — disk health data
Testable drive	Fixed non-removable physical disk eligible for Diag & Repair storage tests, S.M.A.R.T. Analyzer, and CHU storage catalog runs (USB flash and memory cards excluded)
WinPE	Windows Preinstallation Environment — minimal Windows for recovery
WinPE startup log	CDE\Runtime\winpe_startup.log — boot menu → main window handoff trace (also X:\winpe_startup.log in PE)
Work order	Case record on Dashboard tying jobs to a client; internal JSON under CDE\Runtime\WorkOrders\; closed-case PDF under Toolkit\WorkOrders\
Clear Recent	Dashboard control that deletes saved work order JSON in CDE\Runtime\WorkOrders\ (not operator PDFs)
EventKind	Work-order run label: Workflow, ChuPreBoot, CstScan, or FastFix
GuidanceContext	Work-order field storing last symptom id and health preset (Config\SymptomFlows.json)
Suggested path	Dashboard highlight under What are you seeing? — inferred from notes or set by symptom click

Troubleshooting CDE

Problem	Try
Boot menu closes but no main CDE window (text recovery menu)	Sync latest CDE\ToolkitGUI\bin\Release\net8.0-windows\ via Publish-ToolkitGUI.ps1 + SyncUSB.bat; read CDE\Runtime\winpe_startup.log on USB
WinPE window maximize crashes to recovery menu	Fixed 2026-07-03 — update GUI payload on USB; same publish/sync path
Offline OS not detected	Boot menu option 2; check disk cables; load storage drivers if needed
CST finds no targets	Confirm offline Windows loaded; pick correct drive letter
PDF won't open	Install a PDF viewer; check Logs folder for the .pdf file
CHU results empty	Run CHU from rEFInd first; click Refresh in CHU Pre-Boot Results
Boot USB not in storage tests	Expected — removable media (including CDE USB) is excluded from Diag & Repair, S.M.A.R.T., and CHU storage enumeration
Storage tests fail with no drives	Machine may have only removable media visible; attach or select a fixed internal/external HDD
Boot repair greyed out	Boot from USB (WinPE) — not available on live Windows desktop
DISM Warn 0x800f081f	Mount matching Windows ISO; rerun DISM and browse to sources\install.wim, or run Windows Update Repair first
User guide missing	Re-run toolkit publish; confirm HALP\CDE_User_Guide.pdf on USB
GPU Functional Tests unavailable	Confirm CdeGpuDiagCli.exe is beside the published ToolkitGUI.exe; republish the toolkit. WinPE/RDP/basic-display/API limitations may legitimately report N/A
Workflow failed after success (exit code 1)	If clean/report looked OK but GUI shows errors, check CDE\Runtime\Data\workflows\ for internal JSON or Logs\ for the PDF; update toolkit scripts (WorkOrder.ps1 fix 2026-06-26). Re-run the workflow
Temp cleaner many "access denied" warns	Normal for files locked by running apps (IDEs, compilers); review freed space in the summary
DNS flush / proxy reset / icon cache	Use Post-Op Cleanup (removed from Fast Fixes and OS Repair network reset)

Document information

- **Product:** Core Diagnostic Engine (CDE)
- **Launcher:** `Toolkit\CDE.exe`

- **User guide path:** `Toolkit\HALP\CDE_User_Guide.pdf`
- **Regenerate guide:** run `Toolkit\App\Publish-ToolkitGUI.ps1` (canonical publish)